# Omnichannel Retail Sales and Inventory Analytics Dashboard Report

**Group - 7**

Project Team:

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## 1. Introduction

This project focuses on analyzing retail sales data to uncover meaningful insights and support **data-driven decision making**. The analysis covers sales performance across products, cities, channels, and time periods using Power BI.

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## 2. Objectives

- Analyze overall sales performance and key metrics
- Identify top-performing products and categories
- Evaluate sales distribution across cities and channels
- Analyze sales trends over time
- Provide actionable business recommendations

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## 3. Data Preparation

The dataset was cleaned and prepared during the initial phase:

- Removed duplicate records
- Handled missing values
- Standardized date formats
- Verified and corrected data types
- Created calculated fields for analysis

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## 4. SQL Analysis

SQL queries were used to extract key business metrics:

- Total Revenue using SUM ()

- Total Orders using COUNT ()
- Average Order Value using AVG ()
- Product and category performance using GROUP BY
- Revenue distribution by city and channel

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## 5. Dashboard Overview

An interactive Power BI dashboard was developed with:

- KPI cards displaying Total Revenue, Total Orders, and Average Order Value
- Bar charts for category, city, and product performance
- Pie chart for sales distribution by channel
- Line charts for sales trends
- Time-based analysis including:
    - Sales by Day of Week
    - Orders by Day of Week
    - Monthly Sales Trend
- Slicers for filtering by Date, City, Category, and Channel

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## 6. Key Findings

- Sales remain relatively consistent across all weekdays, indicating stable customer demand
- Monthly sales show fluctuations, with stronger performance during mid-year
- Electronics category contributes the highest revenue among all product categories
- Kolkata is the top-performing city in terms of revenue
- Online and Store channels contribute significantly to total sales

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## 7. Recommendations

- Focus marketing campaigns during high-performing months to maximize revenue
- Promote Electronics products as they are the top revenue drivers
- Strengthen business operations in high-performing cities like Kolkata
- Improve strategies for lower-performing categories such as Storage
- Optimize performance across key sales channels

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## 8. Conclusion

The project successfully demonstrates the complete data analytics workflow, including data cleaning, SQL analysis, and dashboard visualization. The insights derived from the dashboard can help businesses make informed decisions and improve overall business performance.

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## 9. Tools Used

- Power BI
- SQL (MySQL)
- Excel
- Python (Pandas)

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### 10. Project Outcome

This project provides a comprehensive analysis of retail sales data and delivers actionable insights to support strategic business decisions.

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## 11. Dataset and Setup Details

**Dataset Description**

The dataset used in this project contains retail sales transactions, including information on orders, products, categories, quantity, price, sales, city, and sales channel.

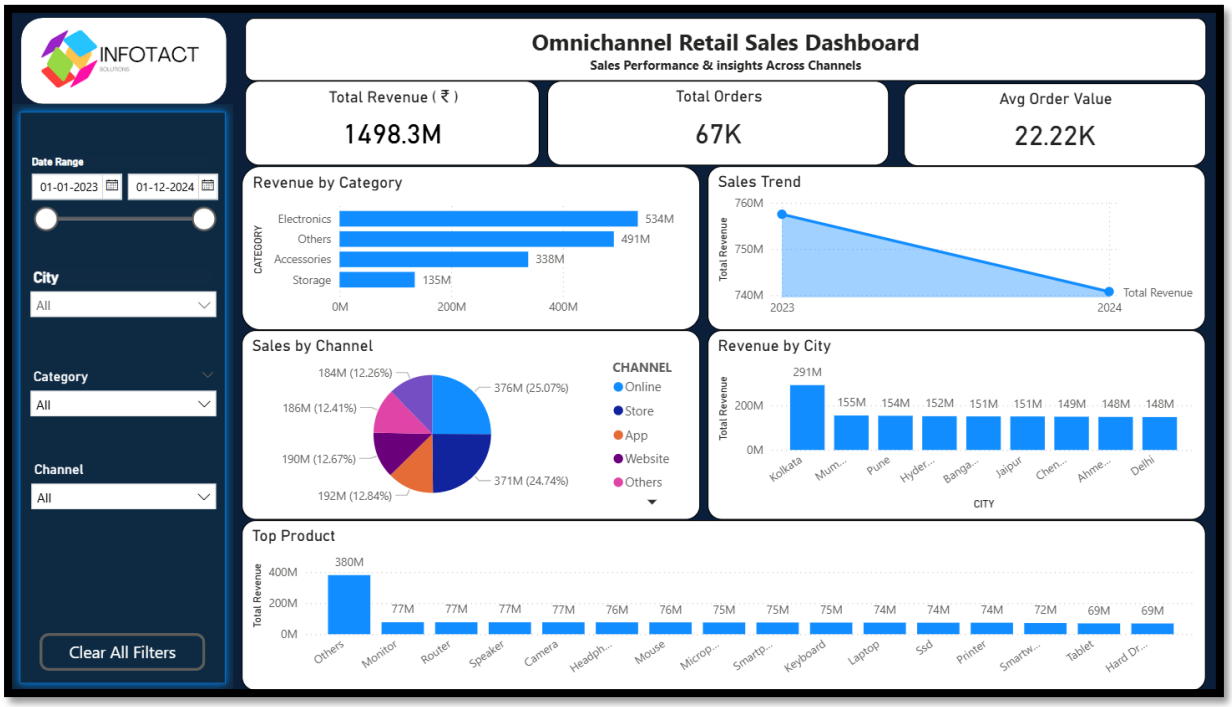
**Setup Instructions**

- The dataset was cleaned and prepared using Excel/Python
- Data was stored and analyzed using MySQL
- SQL queries were used to extract key metrics
- The cleaned data was connected to Power BI for visualization
- Interactive dashboards were created using Power BI features such as charts, slicers, and DAX measures

**Dashboard Screenshots**

The project includes high-resolution screenshots of the Power BI dashboard to illustrate key visualizations and insights.

Figure 1: Sales Overview Dashboard (Page - 1)



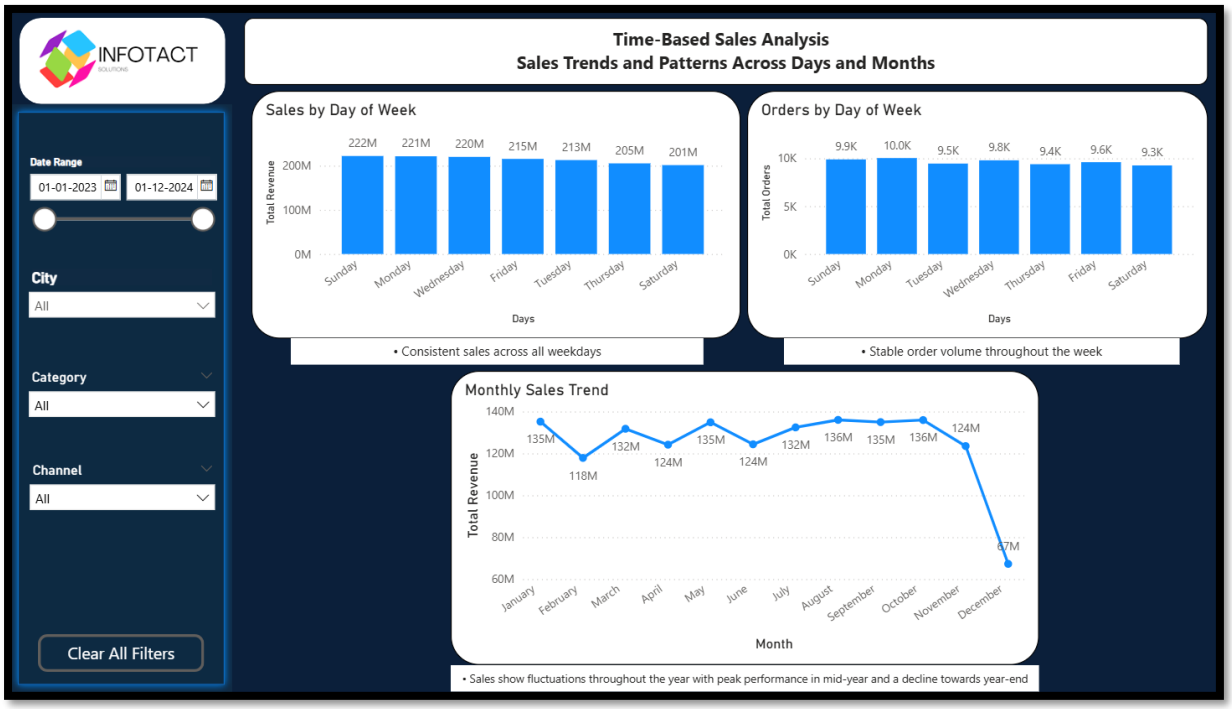


Figure 2: Time-Based Sales Analysis Dashboard (Page – 2)